

| Question number | Answer                                                                                                                                                                                                                                                                           | Additional guidance | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 1(a)            | <p>The only correct answer is A</p> <p><i>B is incorrect because dendrochronology counts tree growth rings</i><br/> <i>C is incorrect because gel electrophoresis separates according to size</i><br/> <i>D is incorrect because polymerase chain reaction amplifies DNA</i></p> |                     | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| (b)(i)          | <p>An answer that includes three, at least one of which is a difference and one a similarity, of the following points:</p> <p>Similarities:</p> <ul style="list-style-type: none"> <li>• all have <math>\beta</math>-carotene (1)</li> <li>• all have chlorophyll a (1)</li> <li>• all have one of the xanthophyll types (1)</li> </ul> <p>Differences:</p> <ul style="list-style-type: none"> <li>• brown seaweed has pigment P and Q that spinach (and red seaweed) does not have (1)</li> <li>• spinach has {three (types of) xanthophylls but both seaweeds have only one / two other (types of) xanthophylls / more types of xanthophyll} (1)</li> <li>• spinach had chlorophyll b which the seaweeds do not have / only spinach has chlorophyll b (1)</li> <li>• spinach has {more types / wider range} of pigments</li> </ul> | <p>DO NOT PIECE TOGETHER<br/>IGNORE bands<br/>plants<br/>explanations of function<br/>of pigments<br/>ACCEPT seaweed for seaweeds<br/>throughout unless contradicted</p> <p>ACCEPT all have xanthophyll</p> <p>ACCEPT only brown seaweed has<br/>pigments P and Q</p> <p>NB both have one of the<br/>xanthophyll types but spinach<br/>has two others = 2 marks<br/>spinach has chlorophyll a and<br/>b whereas the seaweed has only<br/>a = 2 marks</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                 | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)(ii)        | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>• use the distance of pigment P and the {solvent (front) / leading edge} from the origin (1)</li> <li>• (calculate / use) Rf value (1)</li> <li>• look up name of pigment using Rf values for same solvent (1)</li> </ul> | <p>ACCEPT distance moved by pigment P and the {solvent (front) / leading edge}<br/>from a correct equation</p> <p>ACCEPT retention factor / retardation factor / relative (to) front</p> <p>ACCEPT run chromatogram with known pigments {using same solvent / in the same tank}</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                             | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(c)(ii)        | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>to maximise {light / (light) energy} absorption (1)</li> <li>located separately to avoid competition for {(green / blue) light / space} (1)</li> </ul> | <p>ACCEPT brown seaweed has outcompeted red seaweed for {(green / blue) light / space} red/brown below green to avoid competition for {(blue) light / space} two types of red separate to avoid competition for green light</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                  | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(c)(i)         | <p>The only correct answer is C</p> <p><i>A is incorrect because green seaweed reflects green light</i><br/> <i>B is incorrect because green seaweed reflects green light</i><br/> <i>D is incorrect because green seaweed can absorb red light</i></p> | (1)  |

